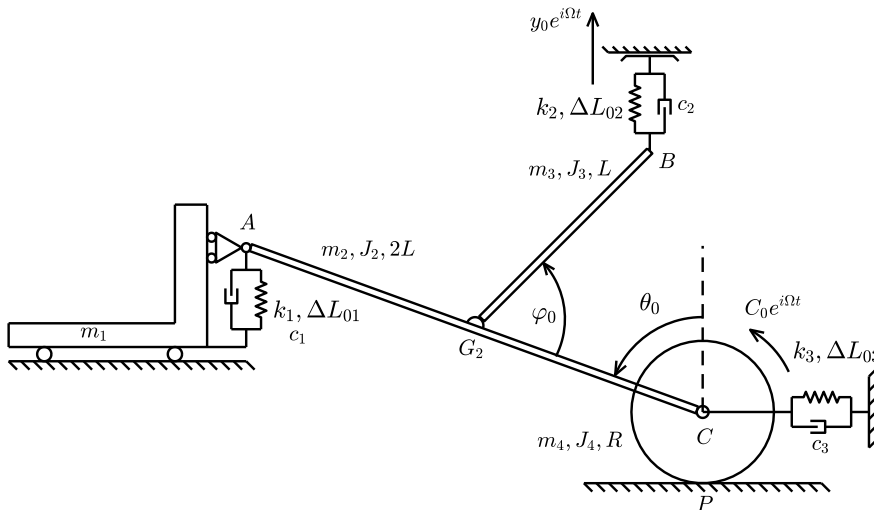


Dynamics of Mechanical Systems

February 01st, 2021



$m_1 = 10 \text{ kg}$	$c_1 = 3 \text{ Ns/m}$
$m_2 = 5 \text{ kg}$	$c_2 = 3 \text{ Ns/m}$
$m_3 = 2.5 \text{ kg}$	$c_3 = 3 \text{ Ns/m}$
$m_4 = 5 \text{ kg}$	$\theta_0 = \pi/3$
$J_2 = 0.4 \text{ kgm}^2$	$\varphi_0 = \pi/3$
$J_3 = 0.05 \text{ kgm}^2$	$x_{01} = 0.04 \text{ m}$
$J_4 = 0.05 \text{ kgm}^2$	$\Omega_1 = 10 \text{ rad/s}$
$L = 0.5 \text{ m}$	$\varphi_{01} = -\pi/3$
$R = 0.15 \text{ m}$	$y_{02} = 0.025 \text{ m}$
$k_1 = 3000 \text{ N/m}$	$\Omega_2 = 30 \text{ rad/s}$
$k_2 = 3000 \text{ N/m}$	$\varphi_{02} = \pi/3$
$k_3 = 3000 \text{ N/m}$	
$\Delta L_{01} = -0.0098 \text{ m}$	
$\Delta L_{02} = 0.0082 \text{ m}$	
$\Delta L_{03} = 0.0 \text{ m}$	

The mechanical system above lies in the vertical plane and is shown in its equilibrium position. The disk rolls without sliding.

The students are requested to:

0. Write on top of each page their surname, name, ID number and SNxxx code (see below)
1. Write the linearized equations of motion around the given equilibrium configuration; in response to this point, upload on MS-Forms one single file in pdf format containing:
 - i. the drawing of the system marked with all kinematic variables used to solve the exam;
 - ii. The vector of independent coordinates chosen to write the system's equations of motion;
 - iii. The Jacobian matrices $[\Delta m]$, $[\Delta c]$, $[\Delta k]$, with evidence of the procedure used to derive these results;
 - iv. The gravitational & second order stiffness, with evidence of the procedure used to derive these results;
2. Compute the natural frequencies and modal shapes;
3. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic torque $C_0 e^{i\Omega t}$. Output: the vertical component of the displacement of point A and the relative rotation between the bars G_2B and AC . Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your matriculation number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic constraint displacement $y_0 e^{i\Omega t}$. Output: the rotation of the disk and the horizontal component of the displacement of the center of mass of the rigid bar G_2B . Save the graphical results as SNxxx3.fig and SNxxx4.fig
5. Suppose to connect the mass m_1 to a hydraulic actuator in order to move it according to the time history $x(t) = x_0 e^{i(\Omega_1 t + \varphi_1)}$. Compute the time history ($T = 2 \text{ s}$, $dt = 0.001 \text{ s}$) of the normal and tangential contact forces in the contact point P applying the constraint displacements $x(t)$ and $y_{02} e^{i(\Omega_2 t + \varphi_2)}$. Save the graphical result as SNxxx5.fig and SNxxx6.fig. In response to this point, also include in the pdf file uploaded on MS-Forms:
 - v. A drawing and the symbolic equations required to derive normal and tangential contact forces in the contact point P.

The answer to points 2 – 5 shall be provided by delivering the MATLAB script and .fig files as a compressed folder uploaded on BeeP, see instructions below.

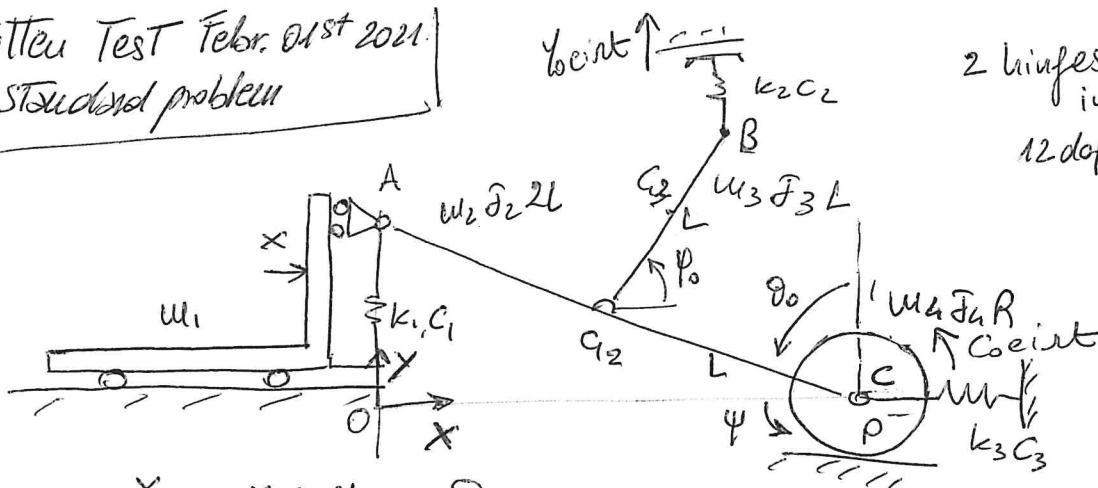
Please be aware that the mark will be lowered by two points if:

- pdf files without the SNxxx code on top of the first page;
- the name of the zip file different from SNxxx.zip;
- the Beep field Title filled not accordingly to the indications given below at point 7.

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

1. Connect to the site <https://beep.metid.polimi.it> and enter the portal using your credentials;
3. Select section **Homework** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder **Test February 01st 2021 - standard**
5. Click on button Add and select Single document;
6. Browse the compressed file SNxxx.zip using the button Browse... (or Sfogliala...)
7. Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
8. Confirm the procedure clicking the button Publish.

Written Test Febr. 01st 2021
Standard problem



2 hinges + 3 carts + contact
in 2D $\Rightarrow$ 9 d.o.f.
 $12 \text{ dof} - 9 \text{ doc} \rightarrow \underline{3 \text{ dof}}$

$$\underline{x} = \begin{Bmatrix} x \\ \theta \\ \phi \end{Bmatrix}$$

$$x_c = x + 2L \sin \theta$$

$$x_{c2} = x + L \sin \theta$$

$$y_A = 2L \cos \theta$$

$$y_{c2} = L \cos \theta$$

$$x_{c3} = x + L \sin \theta + \frac{L}{2} \cos \phi$$

$$y_{c3} = L \cos \theta + \frac{L}{2} \sin \phi$$

$$y_B = L \cos \theta + L \sin \phi$$

$$\psi = \frac{x_c - x_{c0}}{R} = \frac{x + 2L \sin \theta - 2L \sin \theta_0}{R}$$

$$\Delta L_1 = y_A - y_{A0} = 2L \cos \theta - 2L \cos \theta_0$$

$$\Delta L_3 = -x_c + x_{c0} = -x - 2L \sin \theta + 2L \sin \theta_0$$

$$\Delta L_2 = y_B - y_{B0} = y_B - L \cos \theta - L \sin \phi$$

$$W_{PH} = \text{diag}(m_1, m_2 \dot{\theta}_2, m_3 \dot{\theta}_3, m_4 \dot{\theta}_4)$$

$$\underline{F} = \{ C_{eint} \quad F_{el-y} \}^T$$

$$\frac{\partial x_c}{\partial \theta} = 2L \cos \theta \quad \frac{\partial x_{c2}}{\partial \theta} = L \cos \theta \quad \frac{\partial y_A}{\partial \theta} = -2L \sin \theta \quad \frac{\partial y_{c2}}{\partial \theta} = -L \sin \theta$$

$$\frac{\partial x_{c3}}{\partial \theta} = L \cos \theta \quad \frac{\partial x_{c3}}{\partial \phi} = -\frac{L}{2} \sin \phi \quad \frac{\partial y_{c3}}{\partial \theta} = -L \sin \theta \quad \frac{\partial y_{c3}}{\partial \phi} = \frac{L}{2} \cos \phi$$

$$\frac{\partial y_B}{\partial \theta} = -L \sin \theta \quad \frac{\partial y_B}{\partial \phi} = L \cos \phi$$

$$\Delta_{K2} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & L \cos \theta_0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -L \sin \theta_0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & L \cos \theta_0 & -L/2 \sin \phi_0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -L \sin \theta_0 & L/2 \cos \phi_0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 2L \cos \theta_0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1/R & -2L/R \cos \theta_0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} x \\ x_{c2} \\ y_{c2} \\ \theta \\ x_{c3} \\ y_{c3} \\ \phi \\ x_c \\ \psi \end{matrix}$$

$$\Delta_K = \begin{bmatrix} 0 & -2L \sin \theta_0 & 0 \\ 0 & L \sin \theta_0 & -L \cos \phi_0 \\ -1 & -2L \cos \theta_0 & 0 \end{bmatrix}$$

$$\Delta_Q = \begin{bmatrix} -1/R & -2L/R \cos \theta_0 & 0 \\ 0 & -L \sin \theta_0 & L \cos \phi_0 \end{bmatrix}$$

$$\frac{\partial^2 \Delta L_1}{\partial \theta^2} = -2L \cos \vartheta$$

$$[K_{EL, II 1}] = k_1 \Delta L_1 \begin{bmatrix} 0 & 0 & 0 \\ 0 & -2L \cos \vartheta_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial^2 \Delta L_2}{\partial \theta^2} = L \cos \vartheta$$

$$[K_{EL, II 2}] = k_2 \Delta L_2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & L \cos \vartheta_0 & 0 \\ 0 & 0 & L \sin \varphi_0 \end{bmatrix}$$

$$\frac{\partial^2 \Delta L_3}{\partial \theta^2} = 2L \sin \vartheta_0$$

$$[K_{EL, II 3}] = k_3 \Delta L_3 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2L \sin \vartheta_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

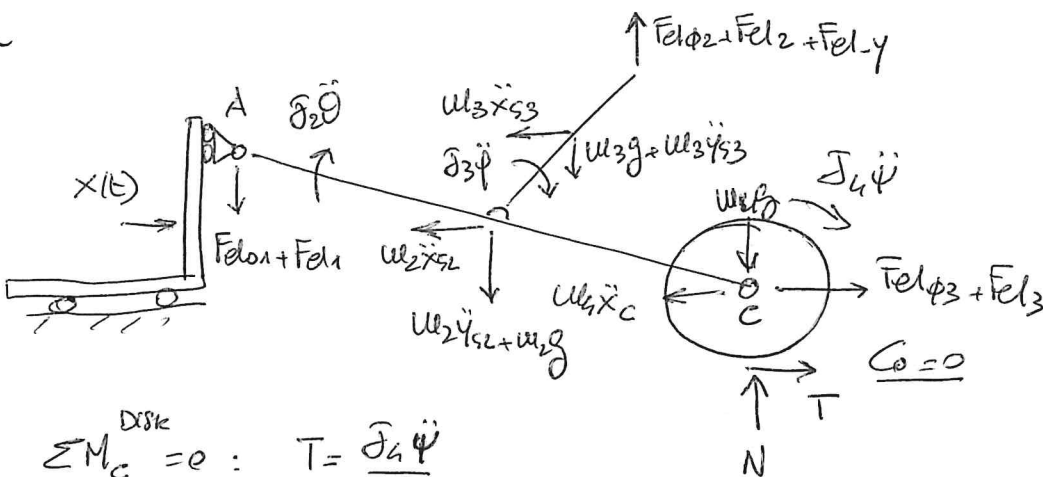
$$\frac{\partial^2 \Delta L_4}{\partial \theta^2} = -L \cos \vartheta$$

$$[K_{s2}] = u_2 g \begin{bmatrix} 0 & 0 & 0 \\ 0 & -L \cos \vartheta_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial^2 \Delta L_5}{\partial \theta^2} = -L \cos \vartheta$$

$$\frac{\partial^2 \Delta L_5}{\partial \varphi^2} = -\frac{L}{2} \sin \varphi$$

$$[K_{s3}] = u_3 g \begin{bmatrix} 0 & 0 & 0 \\ 0 & -L \cos \vartheta_0 & 0 \\ 0 & 0 & -\frac{L}{2} \sin \varphi_0 \end{bmatrix}$$



$$\sum M_C^{Disk} = 0 : T = \frac{J_4 \ddot{\psi}}{R}$$

$$\sum F_y^{AC+GB+Disk} = 0 : N = \underbrace{u_2 g + u_3 g + F_{el\phi 1} - F_{el\phi 2}}_{static} + \underbrace{u_2 \ddot{y}_{s2} + u_3 \ddot{y}_{s3} - F_{el2} - F_{el-y} + F_{el1}}_{dynamic}$$